ECCV 2026 | MALMÖ | SEPT 8–12

2nd Workshop on Marine Vision

MarineMamba: Multi-Granularity CLIP Feature Integration with Dual Mamba for Underwater Classification

Alireza Hoseinzade, Mostafa Shahbazi Dil, Mohammad Mahmoudabadi, Mansoor Rezghi
Tarbiat Modares University, Tehran, Iran

DeepTenLab

Tarbiat Modares University

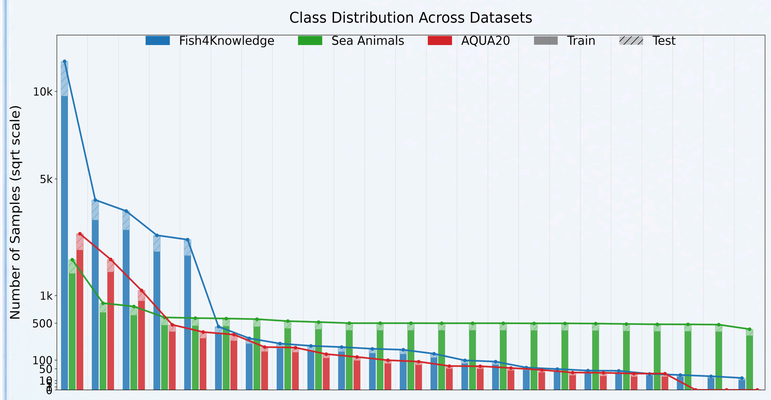
Implementaion

1. Motivation

🌊 Underwater conditions: Color attenuation • scattering • blur

🐟 Fine-grained species: Very similar shape • texture • color

⚠️ Small & imbalanced datasets increases risk of overfitting



➡️ Need: Lightweight, robust underwater classification on:

🐠 Fish4Knowledge — 23-class underwater fishes,

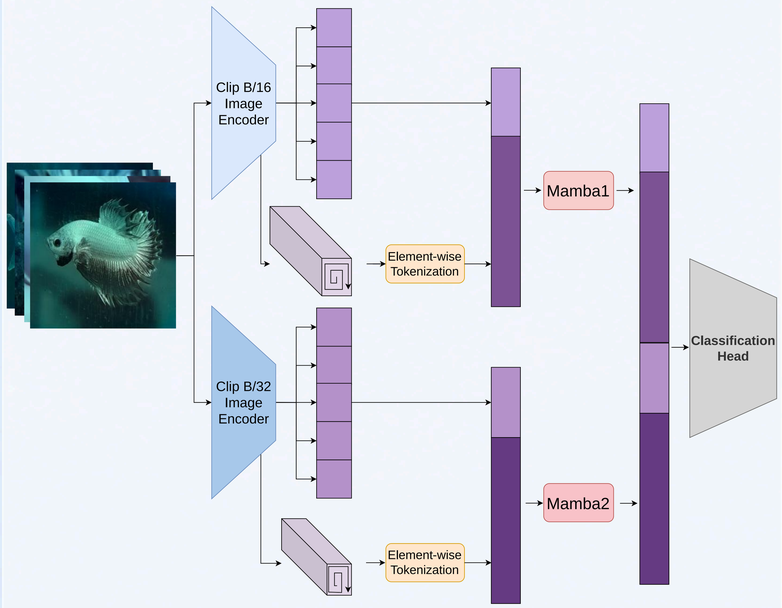
🦈 Sea Animals 23 — 23-class marine animals,

🐟 AQUA20 — 20-class underwater species.



2. Method

MarineMamba combines global context + local details + spatial modeling to achieve efficient underwater species classification.



🧩 Multi-granularity features: Frozen CLIP ViT-B/16 & ViT-B/32 capture fine and coarse visual information.

🌐 Global → Local: Global CLIP features are injected into spatial tokens to connect overall context with local details.

🌀 Spatial modeling: Bidirectional spiral-scan Mamba captures spatial relationships while preserving local continuity.

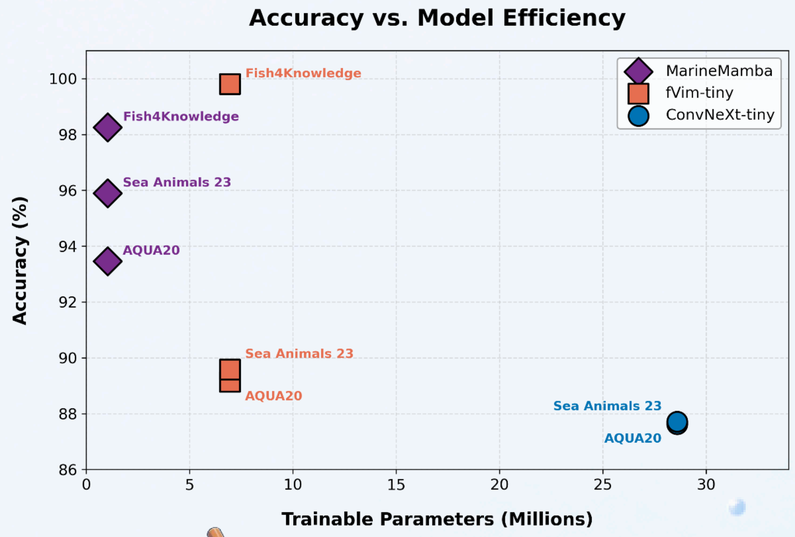
🎯 Feature refinement: Channel and spatial attention emphasize the most informative features.

⚡ Efficient learning: Fine- and coarse-grained features are fused using only 1.03M trainable parameters.

3. Results

🧬 The proposed feature representation produces more compact and better-separated feature clusters, indicating improved discrimination between marine species.

🏆 SOTA Performance with Exceptional Parameter Efficiency





🥇 AQUA20

🥇 Sea Animals 23

🥈 Fish4Knowledge